Review 2

PROJ2999 Capstone Project Introduction

A Deep Learning Approach for Brain Tumor Classification Using MRI Images

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Outline

1. Recap of Review 1
2. Problem Statement and Objectives
3. Requirement Analysis
4. Identification of tools/technology/algorithms
5. Proposed Methodology
6. Project Workflow
7. Data Flow or Use Case Diagram
8. Conclusion
9. Action Plan
10. Disclosure Statements

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Recap of Review 1

Domain

Four-class MRI classification: glioma, meningioma, pituitary tumor and no tumor.

Literature

13 publications reviewed: 11 research papers and 2 surveys, published 2021–2026.

Research gaps

G1 Accuracy shown on one benchmark

G2 Weak classes hidden in the average

G3 No common protocol; cost and clarity ignored

Problem

How well do these models hold up on a second dataset, for each class, and at what cost?

Objectives

Train a transfer-learning classifier, then test it per class on two datasets under one protocol.

Review 1 · 19 Sep 2026
Problem, literature, gaps, objectives

Review 2 · this review
Requirements, tools, methodology, design

Phase 1 closure · Nov 2026
Report, poster, project exhibition

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Feedback from Guide/Reviewers and actions taken after Review–1

Feedback Received in Review 1

- **F1** Three objectives were too broad to verify — reduce them to one or two precise, measurable objectives.
- **F2** Paper numbers on the literature slides did not match the reference list, so citations could not be traced.
- **F3** Datasets, models and the evaluation protocol were still open — fix them before starting system design.

Action taken

- **A1** Objectives merged into two, each with a numeric target and a pass/fail rule. → Slide 6
  O1: macro-F1 ≥ 0.95 with fewer than 10 M parameters · O2: recall ≥ 0.90 for every class, on a second dataset
- **A2** Reference list rebuilt in IEEE order; every bracketed number now resolves to its entry. → Slides 25–26
  18 references numbered by first citation, with the dataset and algorithm sources added
- **A3** Two public datasets, three candidate models and a per-class metric set fixed and documented. → Slides 8–14
  D1: 7,023 images, 4 classes · D2: 3,064 slices from 233 patients · EfficientNet-B0, ResNet-50, custom CNN

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Aim of the Project

Problem Statement

Deep-learning models for brain-tumor MRI report 97–99 % accuracy, but nearly all of it is measured on a single public benchmark and summarised as one overall number [1]–[3]. For four-class classification — glioma, meningioma, pituitary tumor and no tumor — it is not known how such models hold up on a second dataset, which class fails most, or what they cost to run and to explain. This project builds and evaluates a transfer-learning classifier under a documented, reproducible protocol that answers all three.

Objectives addressing the identified research gaps

O1 Build a transfer-learning classifier (EfficientNet-B0, benchmarked against ResNet-50 and a CNN trained from scratch) that reaches **macro-F1 ≥ 0.95** on the untouched test set of the 7,023-image MRI dataset [4] with **fewer than 10 M parameters**.

Addresses G3

O2 Evaluate it under one documented protocol on a second public dataset [5] (3,064 slices, 233 patients, patient-level splits) and report per-class precision, recall, F1, specificity, confusion matrix and Grad-CAM maps, with **recall ≥ 0.90 for every class**.

Addresses G1, G2

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Requirement Analysis

Functional Requirements

- **FR1** **Data ingestion & curation** — load both datasets with labels, check file integrity, remove exact and near-duplicate images.
- **FR2** **Pre-processing** — crop to the brain region, resize to 224 × 224, normalise; augment the training split only.
- **FR3** **Model training** — train EfficientNet-B0, ResNet-50 and a baseline CNN from one config file with fixed seeds.
- **FR4** **Evaluation** — per-class precision, recall, F1, specificity, confusion matrix and ROC-AUC on both datasets.
- **FR5** **Prediction & explanation** — accept one MRI slice; return the class, its confidence and a Grad-CAM heatmap.

Non-functional Requirements

- **NFR1** **Accuracy** — macro-F1 ≥ 0.95 on the held-out test set; recall ≥ 0.90 for every class.
- **NFR2** **Reproducibility** — fixed seeds, frozen split files, logged configs; a rerun lands within ±0.5 points of macro-F1.
- **NFR3** **Efficiency** — under 10 M parameters; ≤ 50 ms per image on a T4 GPU and ≤ 1 s on a laptop CPU.
- **NFR4** **Interpretability** — every prediction carries a heatmap, so a reader can see which region drove it.
- **NFR5** **Ethics & privacy** — public, de-identified data only; every output is labelled “research use, not a diagnosis”.

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Requirement Analysis (contd.)

Users and their needs



Clinician / radiologist (end user)

A second opinion on a single slice in seconds, with a visible reason (heatmap) instead of a bare label.



Research team (developers)

One config-driven pipeline to retrain, compare models and reproduce every reported number.



Guide & reviewers (evaluators)

Per-class and second-dataset results that can be checked against the published work.

Hardware and software requirements

GPU	NVIDIA T4 16 GB (Google Colab / Kaggle) or a local GPU ≥ 8 GB
CPU / RAM	4+ cores, 16 GB RAM
Storage	≈ 10 GB for datasets, splits, checkpoints and logs
Language	Python 3.11
Deep learning	PyTorch 2.x, torchvision
Data & metrics	NumPy, pandas, scikit-learn
Imaging	OpenCV, Albumentations, imagehash
XAI & interface	pytorch-grad-cam, Streamlit
Tooling	Git + GitHub, TensorBoard; Windows 11 / Ubuntu 22.04

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Identification of Tools/Technology/Algorithms

Tools

PyTorch 2 + torchvision [6]
ImageNet-pretrained backbones, mixed-precision GPU training

scikit-learn [7]
audited per-class metrics, confusion matrix, ROC-AUC

OpenCV + Albumentations [8]
brain-region cropping and train-time augmentation

pytorch-grad-cam [9]
class-activation heatmap for every prediction

Streamlit
browser prototype: upload a slice, see class + heatmap

Git / GitHub + TensorBoard
versioned code and configs, training curves

Technology

Transfer learning
reuses ImageNet features [10]; labelled MRI is scarce

Explainable AI (Grad-CAM)
shows the region behind each decision, as in [11]

Cloud GPU (Colab / Kaggle T4)
free, identical environment for all four members

Algorithms [12], [13]

EfficientNet-B0 (primary)
5.3 M params; best accuracy per FLOP [12]

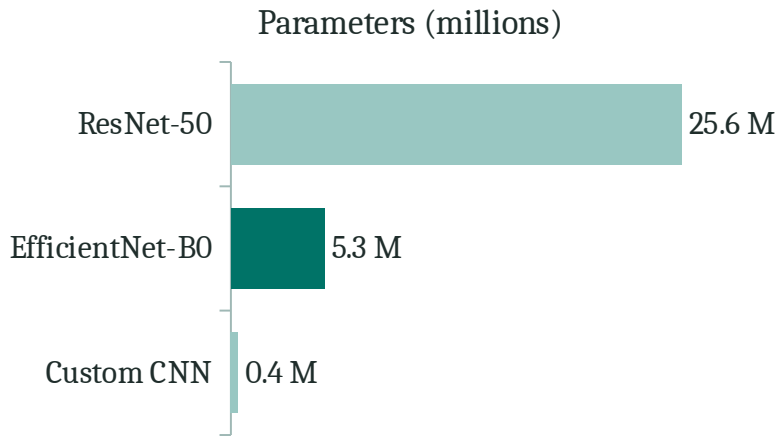
ResNet-50 (baseline)
residual backbone used widely in the literature [13]

Custom 4-block CNN
trained from scratch: what does pre-training add?

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Identification of Algorithms (contd.) — Rationale

Model	Params	GMACs	Init	Role in the study
EfficientNet-B0 [12]	5.3 M	0.39	ImageNet	Primary — compound scaling gives the best accuracy per FLOP; meets NFR3
ResNet-50 [13]	25.6 M	4.1	ImageNet	Baseline — the backbone most reviewed studies build on
Custom CNN	≈ 0.4 M	≈ 0.7	Random	Ablation — measures what pre-training adds on scarce MRI data [17]



Common training recipe (identical for all three models)

- **Two-stage fine-tuning** — 5 epochs on the new head only, then the whole network at a 10× lower learning rate.
- **Loss** — class-weighted cross-entropy with label smoothing 0.1 [15], so the smallest class still counts.
- **Model selection** — on validation data only; the test split is opened once, for the final numbers.
- **Optimiser** — AdamW [14], weight decay 1e-4, batch size 32, mixed precision.
- **Schedule** — cosine learning-rate decay with 1-epoch warm-up [16]; early stopping on validation macro-F1.
- **Explainability** — Grad-CAM [9] on the last convolutional block of each model.

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Proposed Methodology

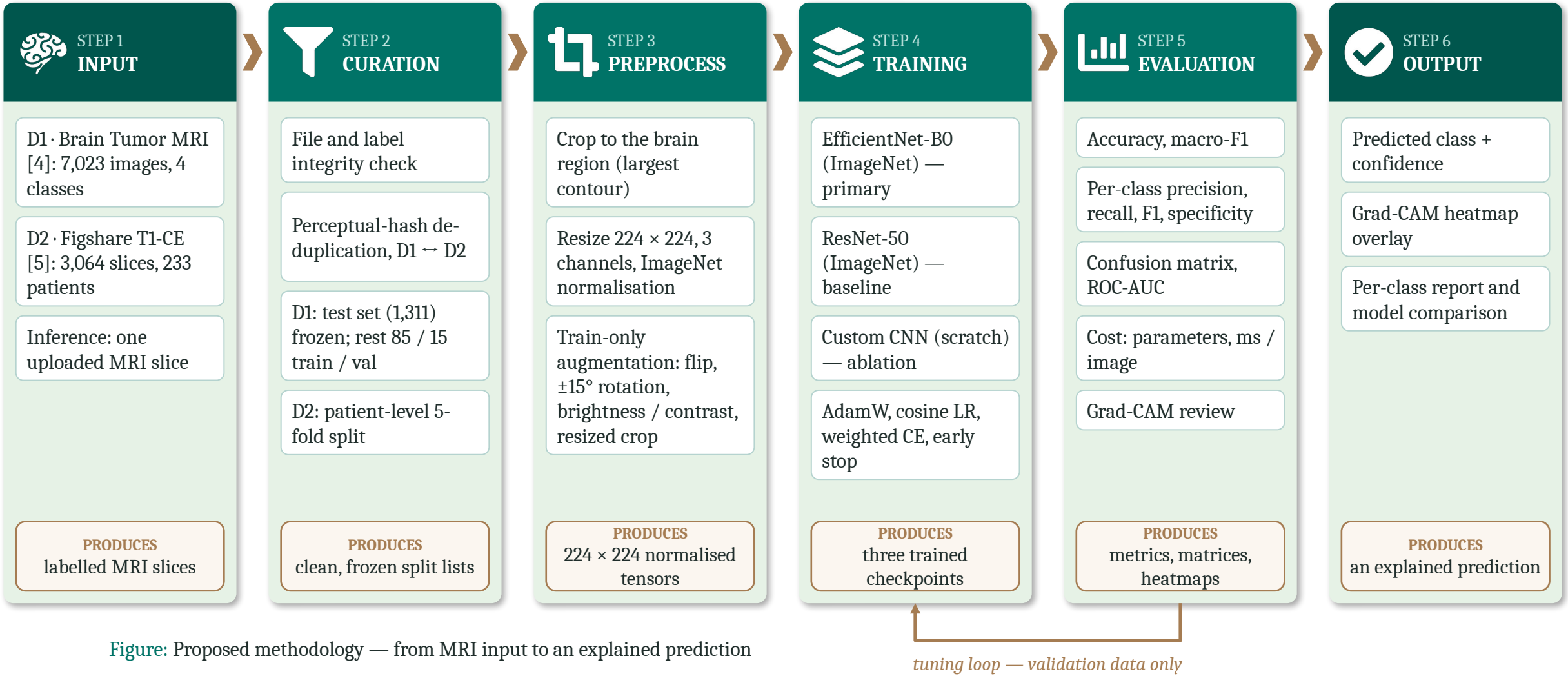


Figure: Proposed methodology — from MRI input to an explained prediction

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Outline

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2. Problem Statement and Objectives
3. Requirement Analysis
4. Identification of tools/technology/algorithms
5. Proposed Methodology
6. Project Workflow
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9. Action Plan
10. Disclosure Statements

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Proposed Workflow

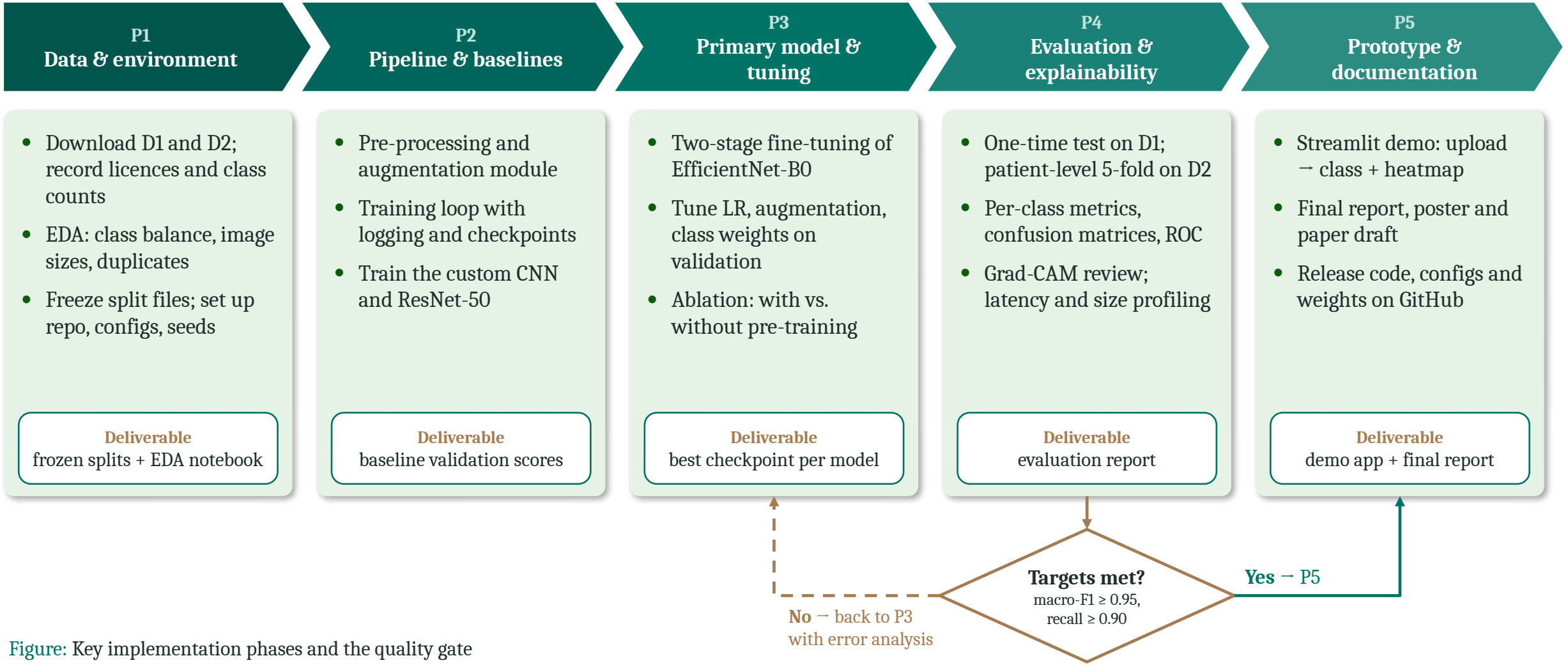


Figure: Key implementation phases and the quality gate

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5. Proposed Methodology
6. Project Workflow
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Data Flow Diagram (DFD) — Level 0 (Context Diagram)

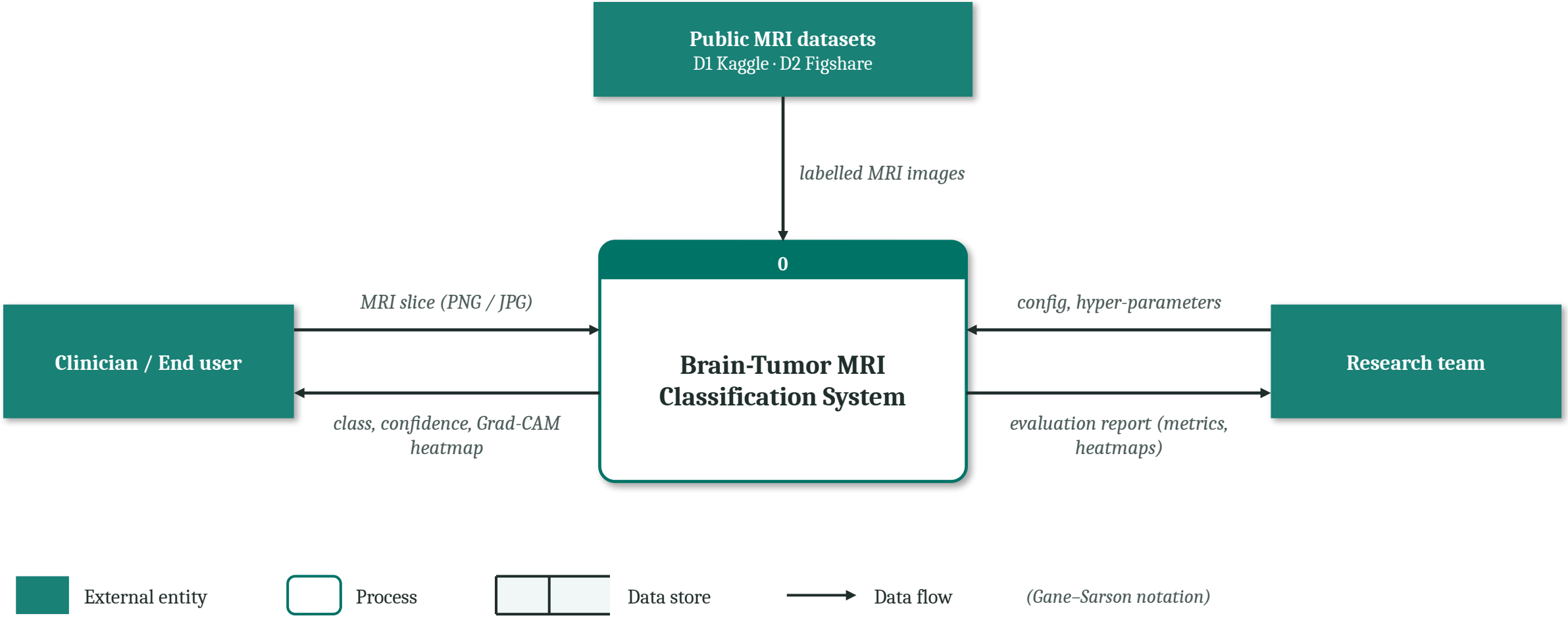


Figure: Level-0 DFD of the proposed system

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Data Flow Diagram (DFD) — Level 1

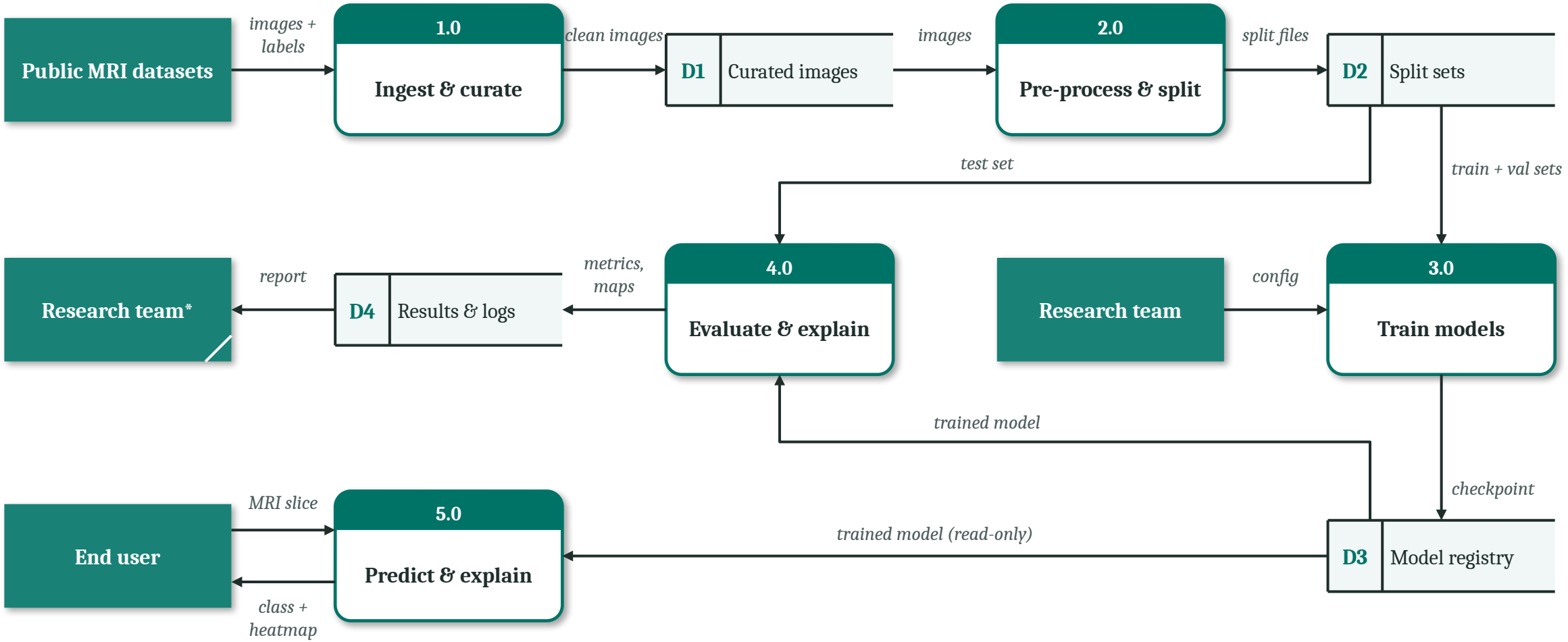


Figure: Level-1 DFD of the proposed system

** duplicate of the same entity, drawn twice to avoid crossing flows*

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Use Case Diagram

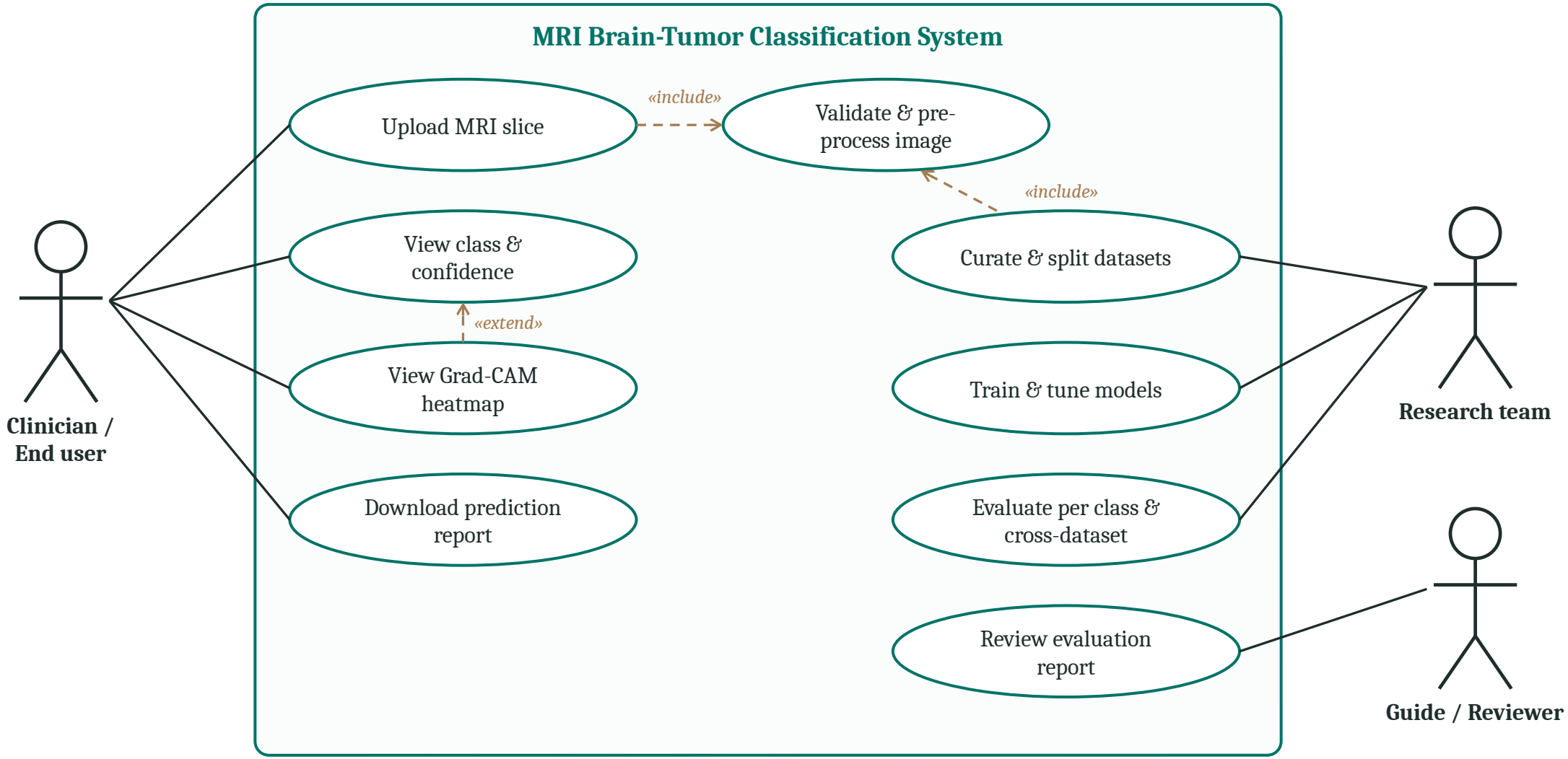


Figure: Use case diagram (UML) of the proposed system

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4. Identification of tools/technology/algorithms
5. Proposed Methodology
6. Project Workflow
7. Data Flow or Use Case Diagram
8. Conclusion
9. Action Plan
10. Disclosure Statements

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Conclusion

We will build a four-class MRI classifier around **EfficientNet-B0**, fine-tuned from ImageNet and benchmarked against **ResNet-50** and a CNN trained from scratch. All three pass through the same pipeline: de-duplicated data, frozen splits, train-only augmentation and validation-only tuning.

Each model is scored **per class on two public datasets**, with Grad-CAM showing where it looked and with its size and latency recorded. The expected outcome is a reproducible account of how well such a classifier works, for which tumor class, and at what cost — the three questions the reviewed literature leaves open.

≥ 0.95

macro-F1 on the untouched D1 test set

≥ 0.90

recall for every class — no hidden weak class

< 10 M

parameters; ≤ 50 ms per image on a T4 GPU

2 datasets

one documented protocol, a heatmap for every prediction

Targets fixed before training — they are pass/fail criteria, not results.

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Workplan and Timeline

Table: Proposed timeline of the project

Task	Task Owner	% of Task Completed	Sept	Oct	Nov
Problem Identification	Chakardhar	100%			
Literature Survey	Nikhil	100%			
Project Objectives	Sahil	100%			
Workplan	Yaswanth	100%			
Requirement Analysis	Sahil	100%			
Identification of Tools/Technology/Algorithms	Nikhil	100%			
System Design (DFD or UMLs)	Yaswanth	100%			
Proposed Methodology	Chakardhar	100%			
Closure of Phase 1/ Project Exhibition	All members	Upcoming			

Sept — Review 1 scope Oct — Review 2 scope Nov — Phase 1 closure

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References (1/2)

References in the IEEE format as follows:

[1] M. Aamir et al., "A deep learning approach for brain tumor classification using MRI images," *Computers and Electrical Engineering*, vol. 101, art. no. 108105, 2022, doi: 10.1016/j.compeleceng.2022.108105.

[2] F. J. Díaz-Pernas, M. Martínez-Zarzuela, M. Antón-Rodríguez, and D. González-Ortega, "A deep learning approach for brain tumor classification and segmentation using a multiscale convolutional neural network," *Healthcare*, vol. 9, no. 2, art. no. 153, 2021, doi: 10.3390/healthcare9020153.

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[4] M. Nickparvar, "Brain tumor MRI dataset," Kaggle, 2021, doi: 10.34740/KAGGLE/DSV/2645886.

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[6] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019, pp. 8024–8035.

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[10] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2009, pp. 248–255, doi: 10.1109/CVPR.2009.5206848.

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[14] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.

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[17] V. Vadde and W. Bukaita, "Comparative evaluation of CNN and ResNet18 architectures for MRI-based brain tumor classification using deep learning," *Medical Research Archives*, vol. 13, no. 12, 2025, doi: 10.18103/mra.v13i12.7100.

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9. Action Plan
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Disclosure Statements

- I will maintain academic honesty by doing my own work and avoiding plagiarism, cheating, unauthorized collaboration, and falsifying information, use of AI Content.
- I will properly acknowledge and cite all sources I use, including information from the internet, books, software, and AI tools.
- I will use institutional resources responsibly and only for authorized academic or research activities. I understand that I am responsible for my own actions and decisions.

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